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Stochastic On Time Arrival Problem

Milestone-04

S1-G8-ITS

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CSE 400: Fundamentals of Probability in Computing

Project Context and Motivation

- Intelligent Transportation System
- Goal: Maximize $P(T \leq D)$
- Travel time is random
- Real-world traffic is uncertain
- Need decision under uncertainty

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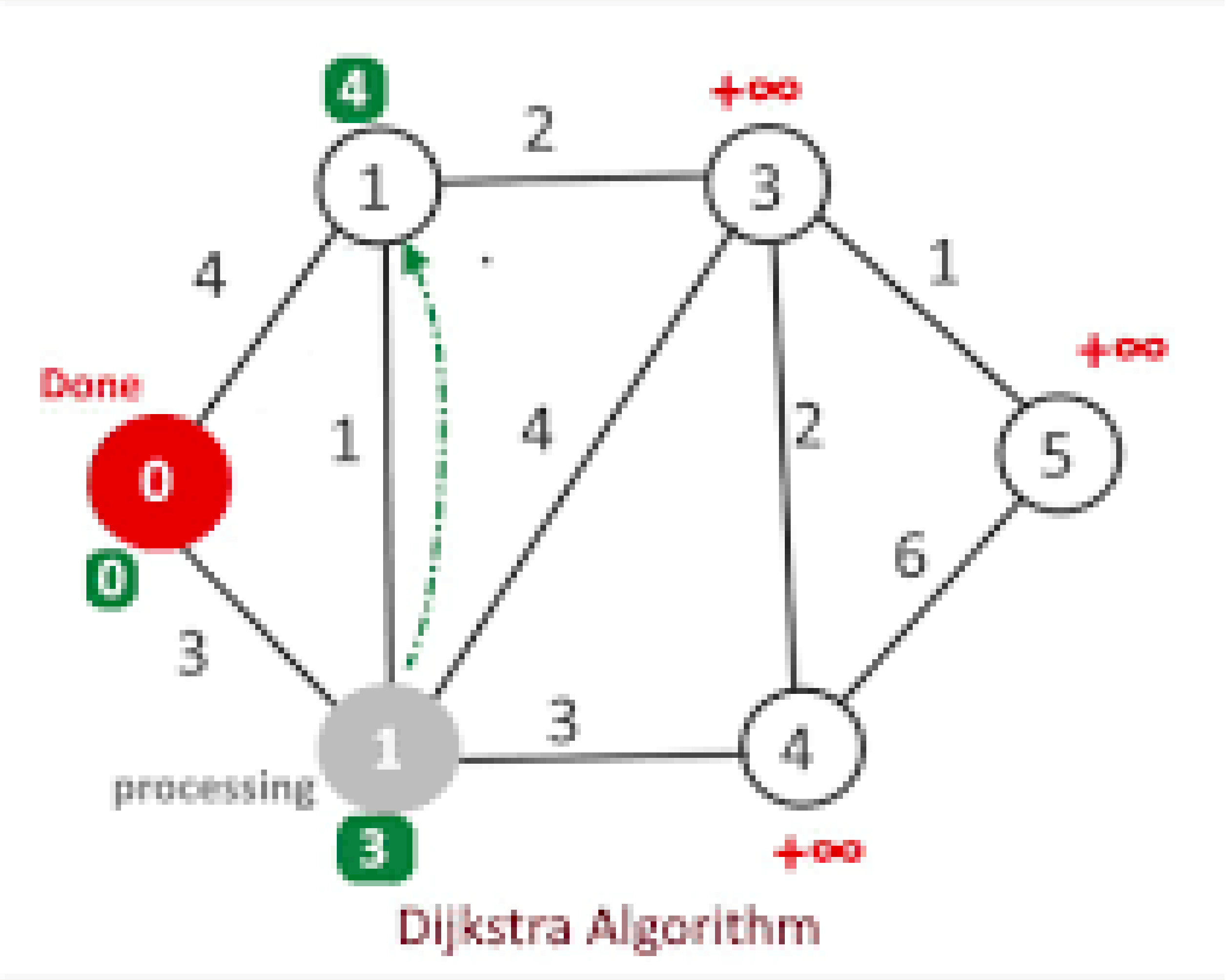
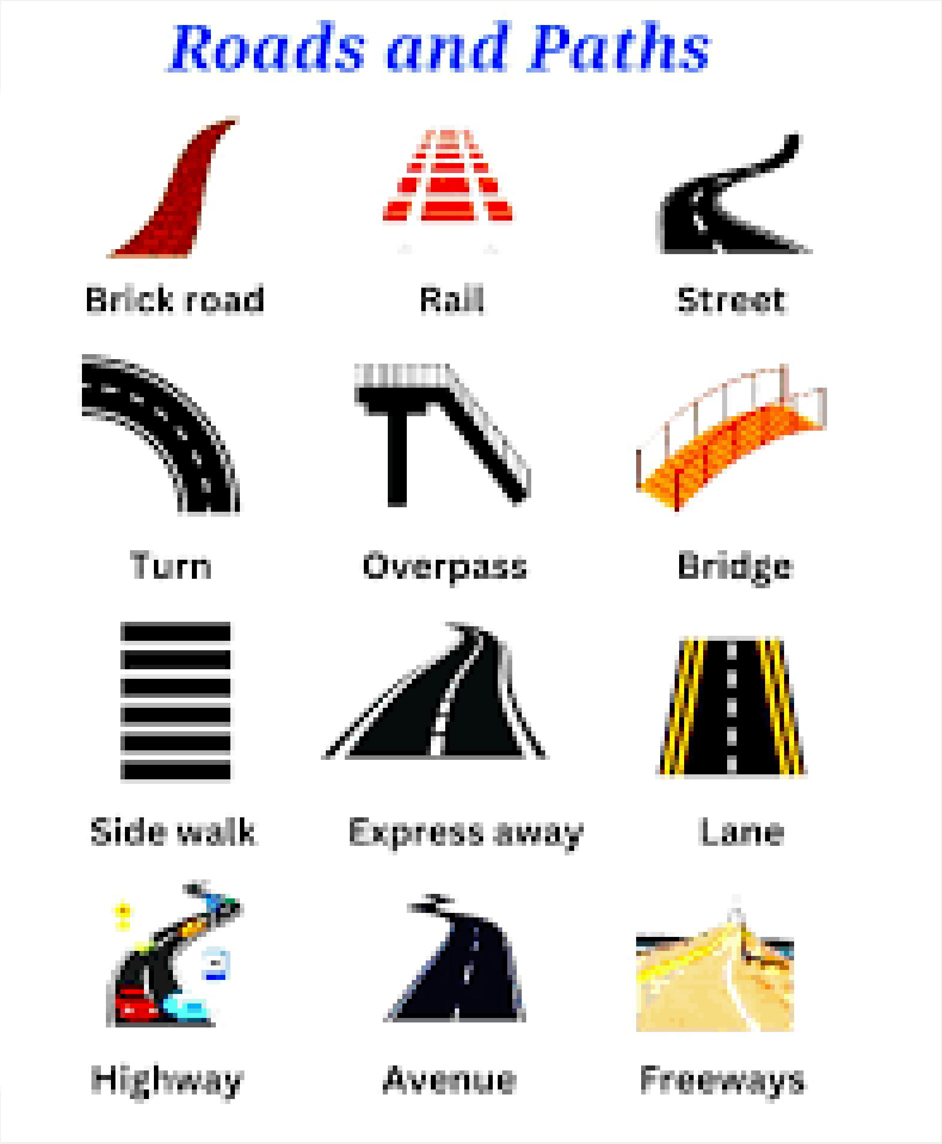
Deterministic Baseline Overview

- Dijkstra Algorithm (LET)
- Uses expected travel time
- Converts PMF → single value
- Finds shortest path
- Ignores uncertainty
- Not always reliable

	Discrete Random Variable	Continuous Random Variable
Mean (Expected Value)	$\mu = E(X) = \sum_{i=1}^{n} x f(x)$	$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$
Variance	$\sigma^2 = V(X) = \sum_{i=1}^{n} (x-\mu)^2 f(x)$	$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$
Standard Deviation (Standard Error)	$\sigma = \sqrt{\sigma^2}$	$\sigma = \sqrt{\sigma^2}$





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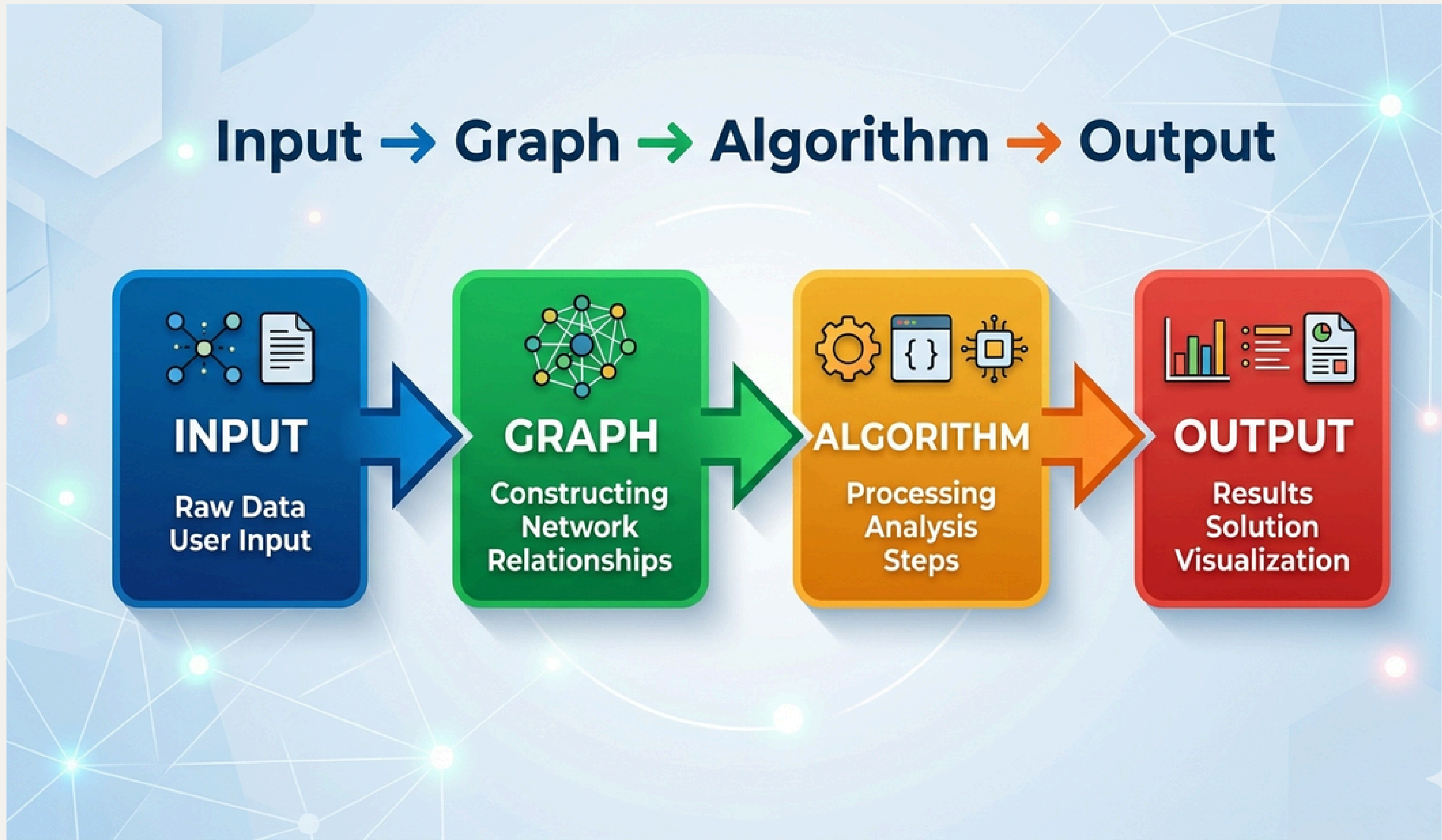
System Pipeline Overview

- Input: Graph + PMF (JSON)
- Graph construction
- Routing using Dijkstra
- Path generation
- Compute $P(T \leq D)$
- Output: path + reliability



Sources of Uncertainty

- Traffic congestion
- Signal delays
- Accidents
- Weather conditions
- Travel time varies
- Deterministic fails here

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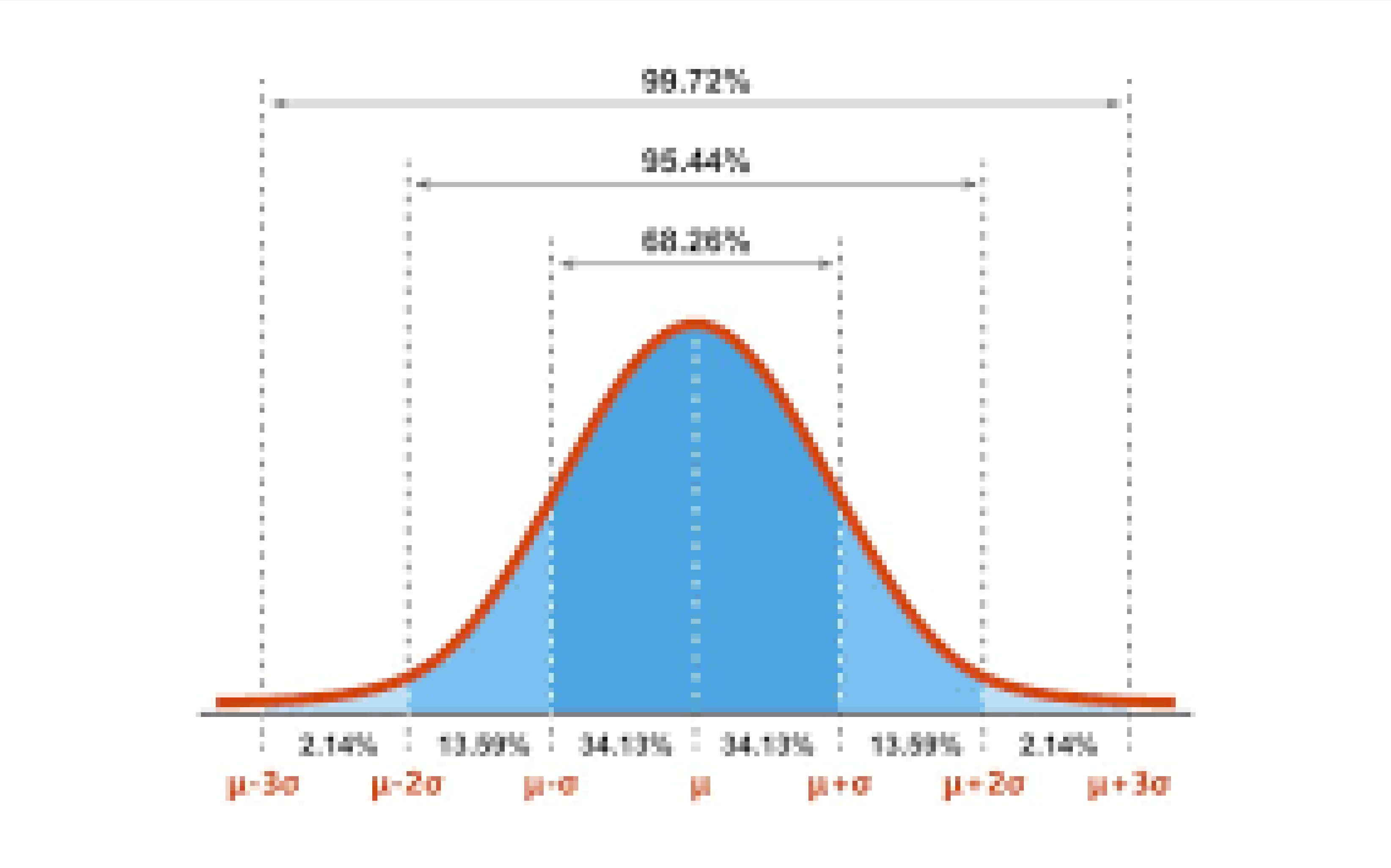
Random Variables in the System

- Travel time = random variable
- PMF for each edge
- Shows probabilities
- Independence assumption
- Gaussian, Uniform, Bimodal
- Models real-world traffic

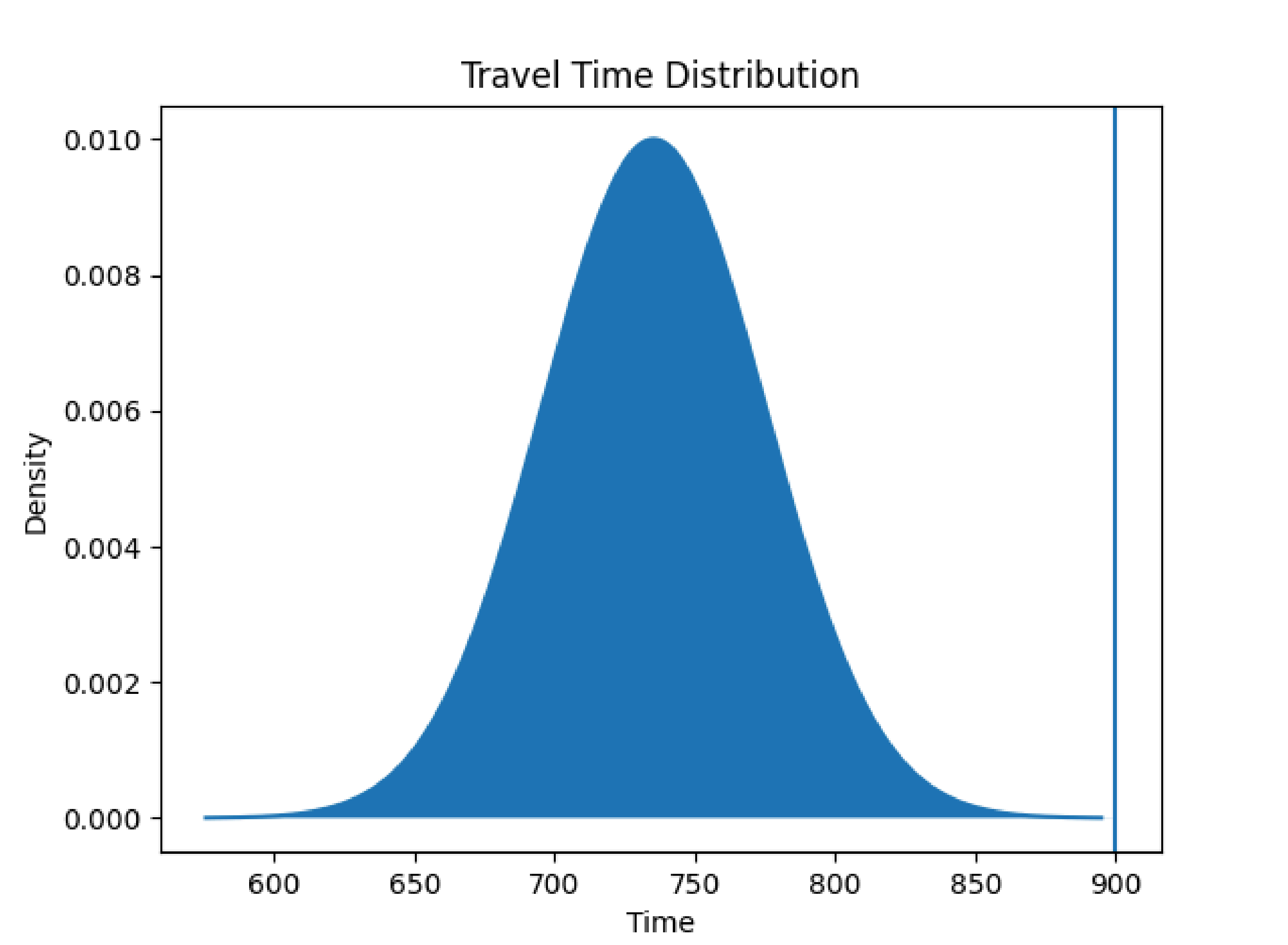


Randomized Algorithm Design

- Randomized Decision Rule: An A^* heuristic approach evaluating true path reliability.
- Introducing Randomness: Edge costs are no longer scalars; the algorithm pulls the full PMF array from the graph data.
- Algorithm Flow: At each node, it calculates the PMF of the path so far, checks the remaining time budget, and prioritizes paths with the highest cumulative probability of success.

Probabilistic Reasoning Behind the Algorithm

- Why it Improves Performance: The shortest path is not always the most beneficial in the real world.
- Conceptual Strategy: By prioritizing reliability over raw distance, the system actively navigates around high-variance, risky edges.
- Mathematical Explanation: The algorithm assumes edges are independent distributions and uses convolution to mathematically determine the exact probability density of the total path.



Implementation of the Randomized Strategy

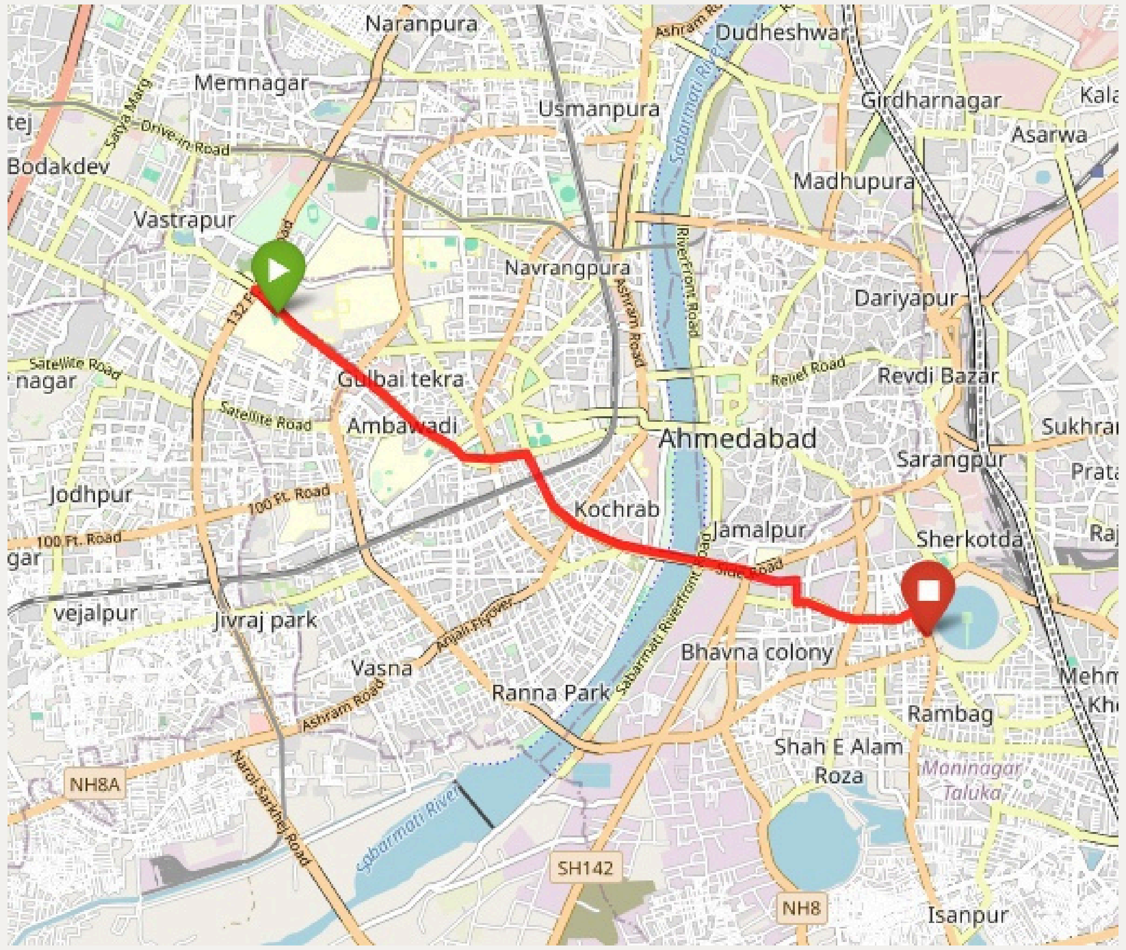
- Integration: Replaced standard weight additions with array-based mathematical convolutions inside the search loop.
- Code Structure: * Uses a priority queue (heapq) pushing state tuples: (-probability, expected_time, node, current_pmf, path).
- Features a custom heuristic function checking the minimum possible time against the remaining budget to prune dead-end branches.

Evaluation Metrics

- Primary Metric 1: Path Reliability / Probability of Success (P(T<=D)).
- Primary Metric 2: Careful physical analysis of the chosen path network.
- Secondary Metric: Algorithmic execution time.
- Why Appropriate: These metrics evaluate the quality and safety of the result in a stochastic environment, rather than just the computational speed of the machine.

Comparative Results

- Deterministic vs Randomized: Tested on a 100-node graph with a strict budget of 550 seconds.
- Visual/Statistical Summary: * Deterministic (LET) chooses a direct route but frequently fails the strict deadline due to high-variance edges.
- Randomized (A^*) chooses a slightly longer, but statistically tighter route, yielding a significantly higher success probability.



Final selected path
with Randomized approach

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Key Insights

- Answer Quality: Randomized algorithms provide unparalleled efficiency regarding the quality and reliability of the route.
- Scalability Challenges: Time complexity for convolution-based search grows much faster than standard Dijkstra on graphs >100 nodes.
- Final Conclusion: Deterministic algorithms provide fast, "good enough" baseline answers, but randomized approaches are mandatory for great, realistic solutions where failure is heavily penalized.

References and Source

Problem Reference

Niknami, M. and Samaranayake, S. (2016). Tractable Pathfinding for the Stochastic on-time Arrival Problem. In: Springer. pp.231–245.

Our Repository

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